

Measuring & Mitigating Popularity Bias in Music Recommender Systems

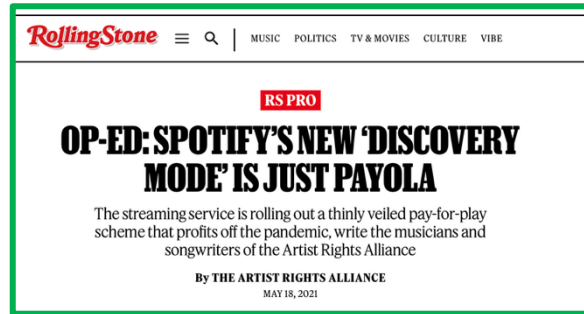
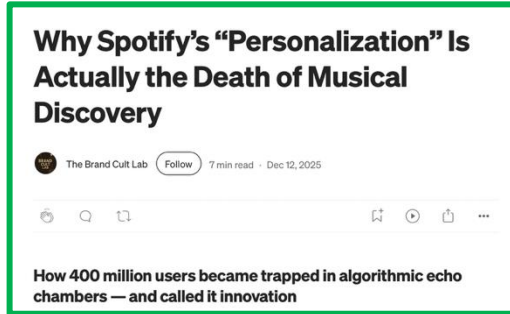


DATA SCIENCE & ARTIFICIAL INTELLIGENCE · PROJECT PROPOSAL

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Why Does Popularity Bias Matter?



Societal & Industrial Perspective

Economic harm to artists: Niche and emerging musicians lose revenue due to suppressed visibility.

Cultural homogenization: Algorithmic monocultures limit users musical exploration and diversity.

Platform trust: Listeners expect personalization, not popularity repetition. Bias erodes trust.

- 1) The Brand Cult Lab. "Why Spotify's 'Personalization' Is Actually the Death of Musical Discovery." Medium. <https://medium.com/@thebrandcult.lab/why-spotifys-personalization-is-actually-the-death-of-musical-discovery-830173509462>
- 2) Artist Rights Alliance. "Op-Ed: Spotify's New 'Discovery Mode' Is Just Payola." Rolling Stone, 2021. <https://www.rollingstone.com/pro/music-biz-commentary/spotify-payola-artist-rights-alliance-1170544>
- 3) D'Souza, S. "The music industry is engineering artist popularity — listeners are right to be angry." The Guardian, 29 Jul 2024. <https://www.theguardian.com/music/article/2024/jul/29/the-music-industry-is-engineering-artist-popularity-and-listeners-are-right-to-be-angry>

Why Does Popularity Bias Matter?

400M+₁

Users on Spotify affected by
algorithmic recommendations

< 20%₂

Of artists generate 80%+ of all
streams (extreme concentration)

70%₃

Revenue share goes to already-
popular artists via algorithmic
amplification

1) Spotify (2024). Loud & Clear: Annual transparency report on the streaming economy. spotify.com/loudandclear

2) Celma, O. (2010). *Music Recommendation and Discovery*. Springer. | Spotify Loud & Clear (2024).

3) Ferraro, A. et al. (2021). *Break the Loop: Gender Imbalance in Music Recommenders*. ACM UMAP 2021.

Literature Synthesis

Paper / Source	Focus Area	Key Contribution	Bias Type	Relevance to Us
Merton (1968) Matthew Effect	Sociology	"Rich get richer" — foundational theory behind popularity amplification in networked systems.	Popularity	<i>Theoretical framing for our bias model</i>
Park & Tuzhilin (2008)	Long-tail RS	First RS paper on long-tail promotion; showed CF systematically under-recommends niche items.	Popularity	<i>Motivates long-tail exposure metrics</i>
Steck (2018)	Calibration / Fairness	Calibrated recommendations to align suggestion distributions with actual user interest profiles.	Popularity	<i>Calibration as a mitigation baseline</i>
Abdollahpouri et al. (2019)	Popularity Bias in RS	Defined ARP & ACLT metrics; showed CF compounds popularity bias across user groups. (FLAIRS 2019)	Popularity	<i>Core metrics we adopt (ARP, ACLT)</i>
Ferraro et al. (2021)	Music RS Fairness	Artist-side fairness in music streaming; CF amplifies mainstream artists disproportionately. (UMAP 2021)	Popularity	<i>Closest domain — direct benchmark</i>
Bird et al. (2020) Fairlearn	Mitigation Toolkit	Open-source toolkit for group fairness constraints (reductions approach) in classification tasks.	Demographic	<i>Gap: no popularity / RS bias support</i>
Bellamy et al. AI Fairness 360	Mitigation Toolkit	IBM toolkit — 70+ metrics & algorithms focused on classification and demographic fairness.	Demographic	<i>Gap: not applicable to item-level RS</i>
This paper (2026)	Music RS Popularity Bias	First study to combine ARP/ACLT measurement + mitigation comparison on real music data (Last.fm / LFM-1b) — filling the gap left by demographic-only toolkits.	Popularity	<i>Novel contribution to the field</i>

What Did We Plan to Explore?

How can **popularity bias** in **music recommender systems** be reliably **measured**, and what **mitigation strategies** most effectively **improve fairness** without sacrificing user satisfaction?

RQ1

Measurement

Which metrics (ARP, ACLT, popularity lift) best capture the extent of popularity bias across different RS algorithms (CF, MF, neural)?

RQ2

Mitigation

Do re-ranking, regularization, or adversarial training approaches reduce popularity bias while preserving recommendation quality?

RQ3

Trade-offs

How do mitigation strategies affect long-tail artist exposure, serendipity, and user-perceived relevance in music datasets?

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H1 Popularity bias exists in raw data and all three recommenders amplify it above the raw-data baseline.

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5-Stage Pipeline



DATASETS

LFM-1b

1B+ listening events · 3,000 users · 352,805 artists
Official Kowald mainstreamness groups. Primary dataset.

XITE

23M sessions · 1M pseudo-users · 23,137 artists
Music video platform 2023-24. Cross-dataset validation.

BIAS METRICS

ARP:

$$(1/|U|) \sum_u (1/|L_u|) \sum_i \phi(i)$$

Higher = more mainstream bias

ACLT:

$$(1/|U|) \sum_u (\sum_i 1[i \in \Gamma]) / |\Gamma|$$

Lower = worse fairness

Gini:

$$\sum_i \sum_j |x_i - x_j| / (2n^2 \cdot x)$$

0=equal, 1=monopoly

RESULTS · RQ1 — ARTIST-SIDE BIAS

Long-tail: Top 20% of Artists Capture 96% of All Plays

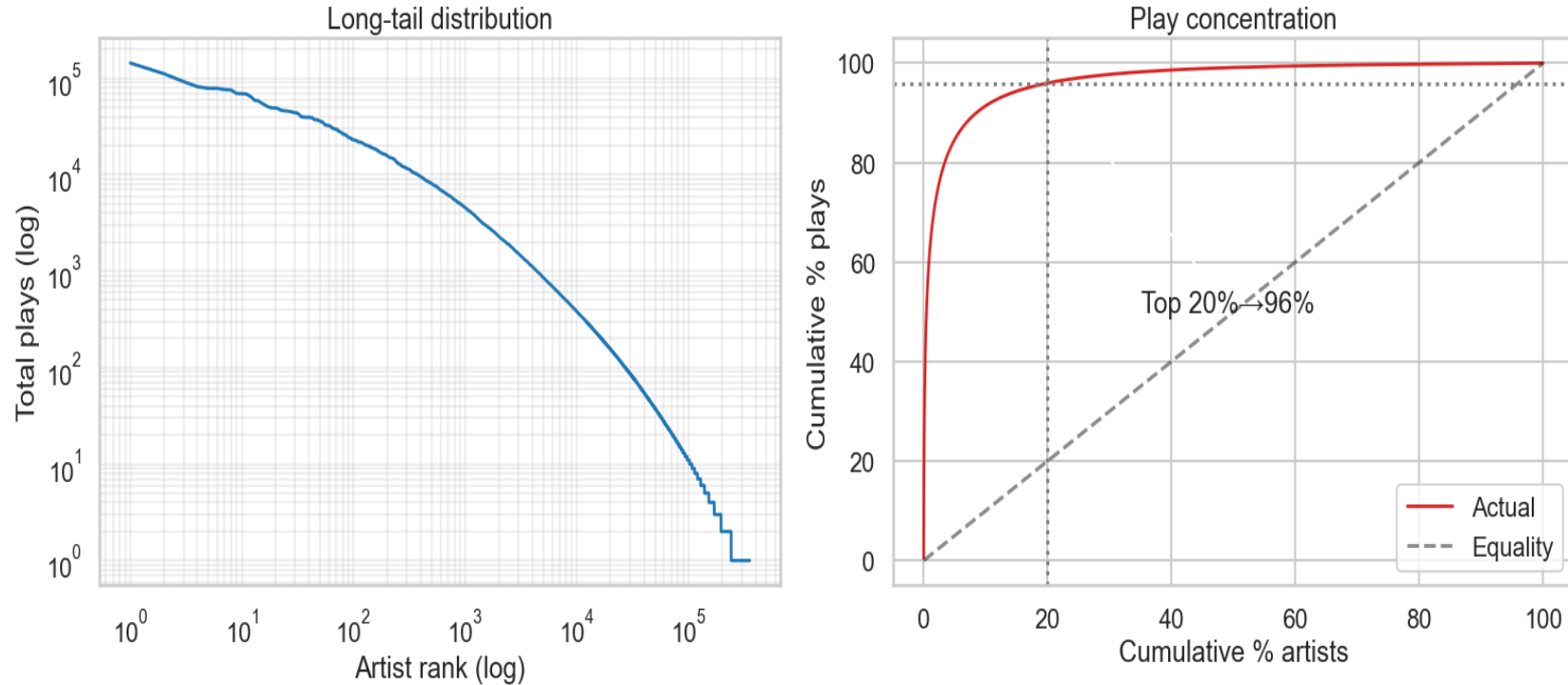


Fig 1. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section 3: Popularity distribution & Lorenz curve

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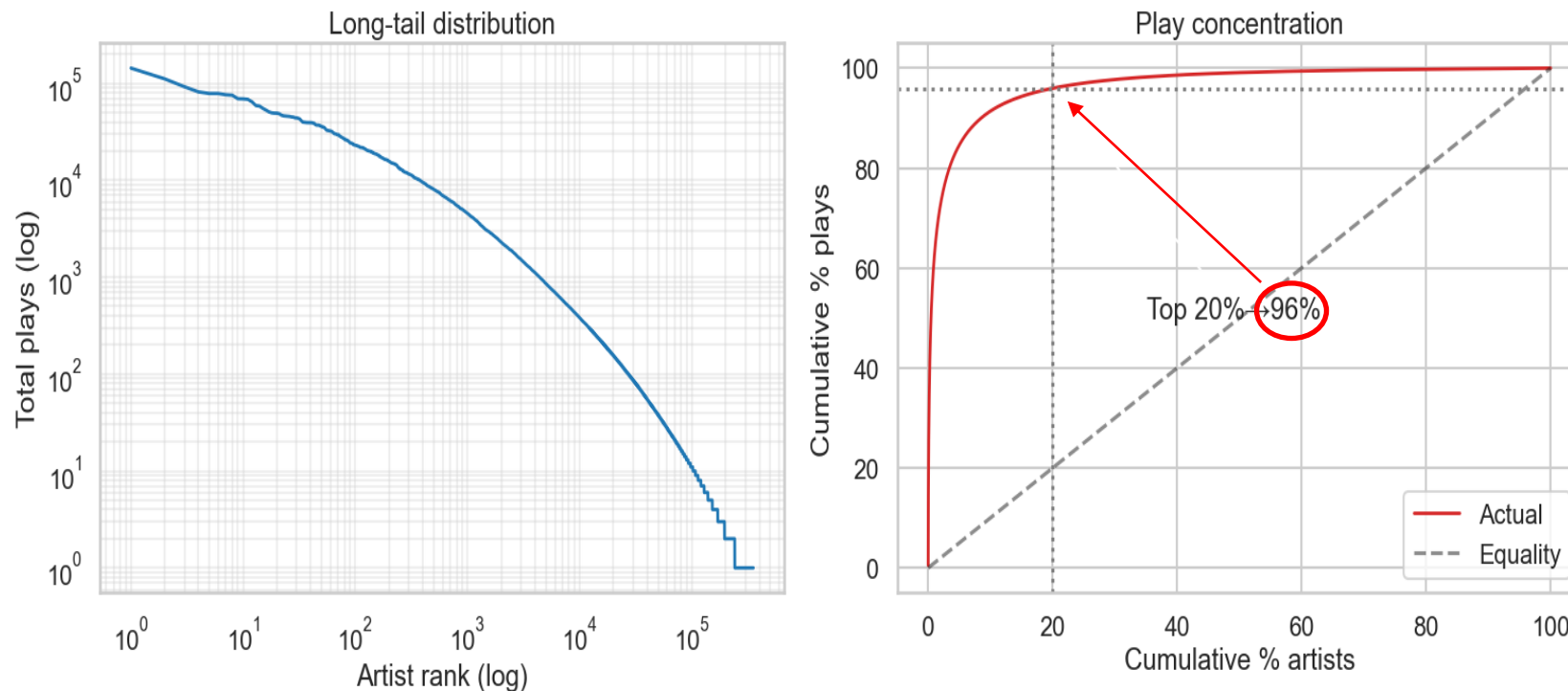


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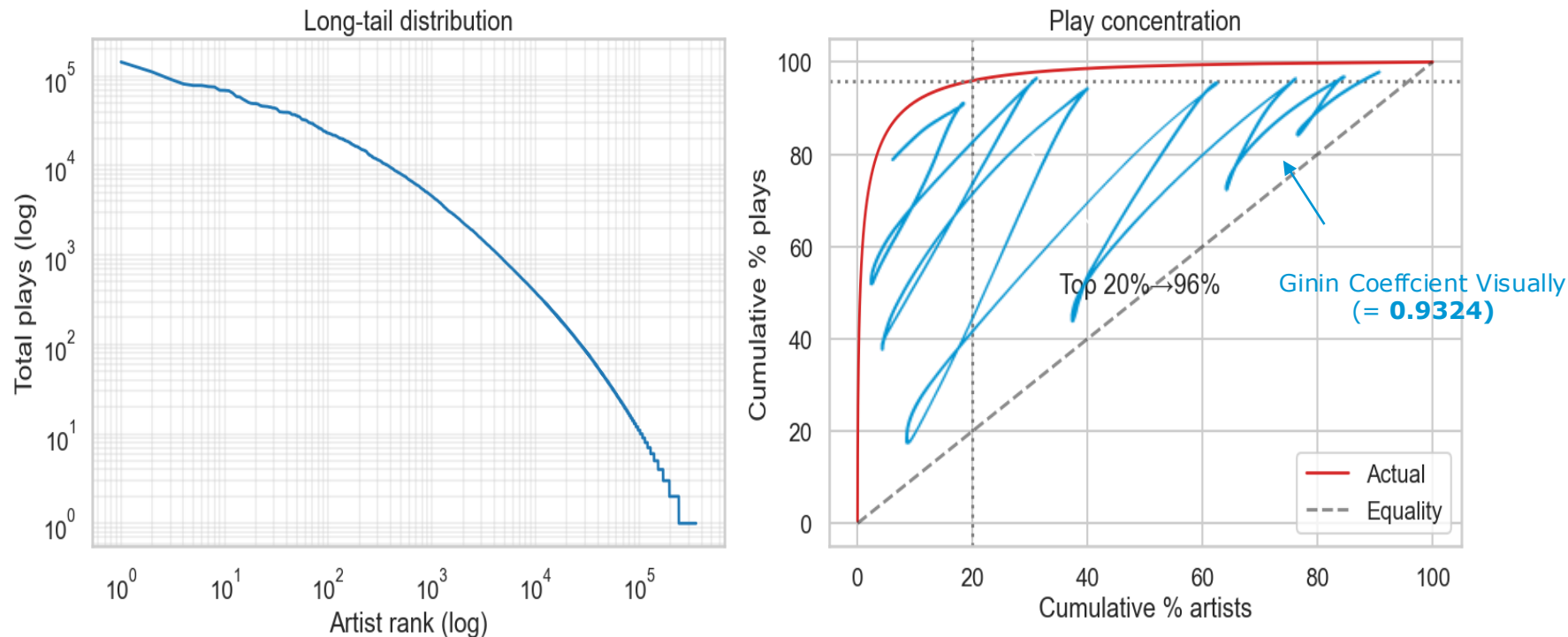


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Tail Artists Are Nearly Invisible

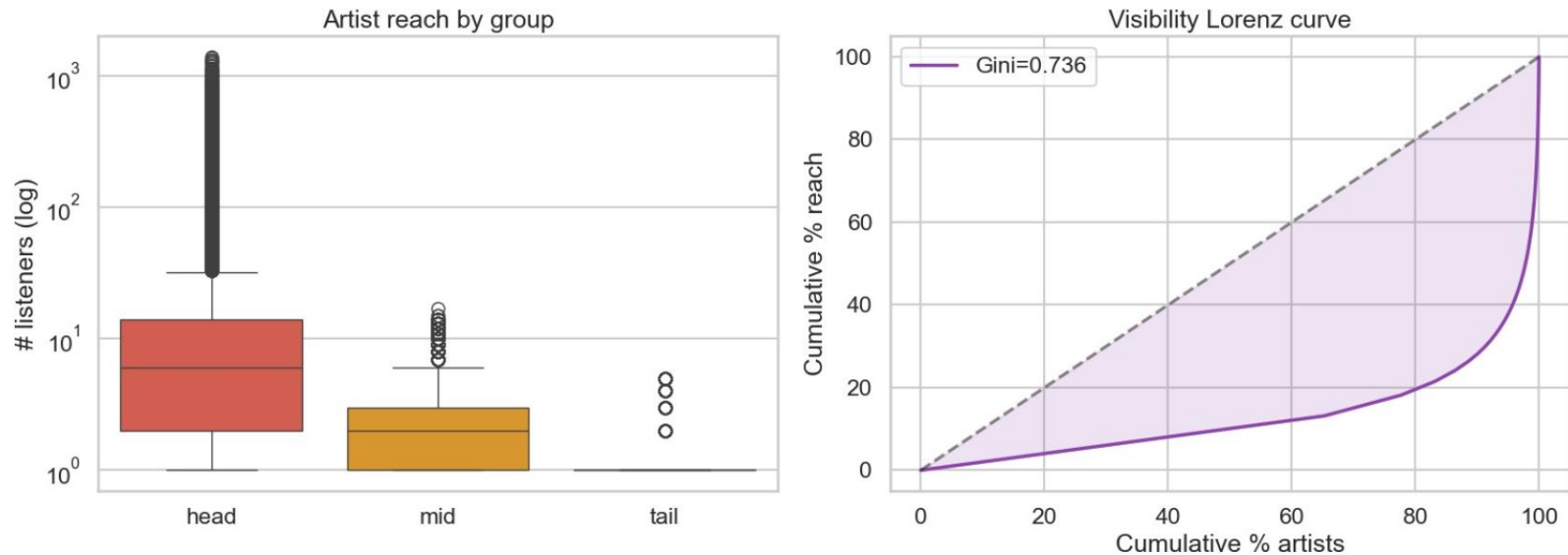


Fig 2. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section 4: Head / Mid / Tail groups & reach inequality

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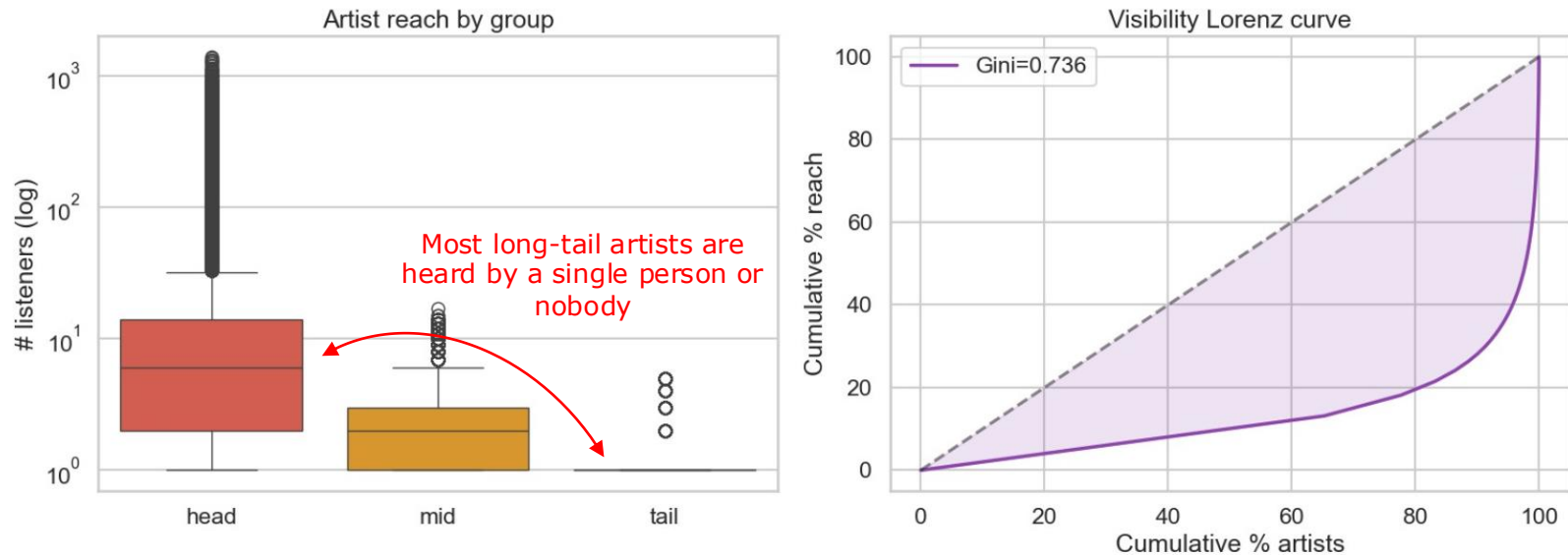


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RESULTS · RQ1 — USER-SIDE BIAS

High-Mainstreaminess Users Consume More Popular Artists

User-side popularity bias (raw data)

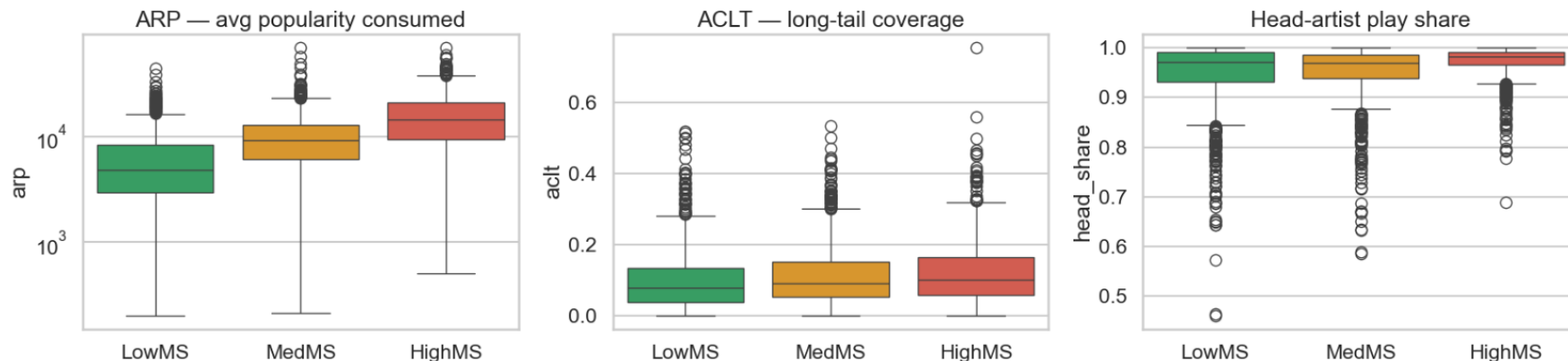


Fig 3. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section 5: Per-user ARP / ACLT / head-share

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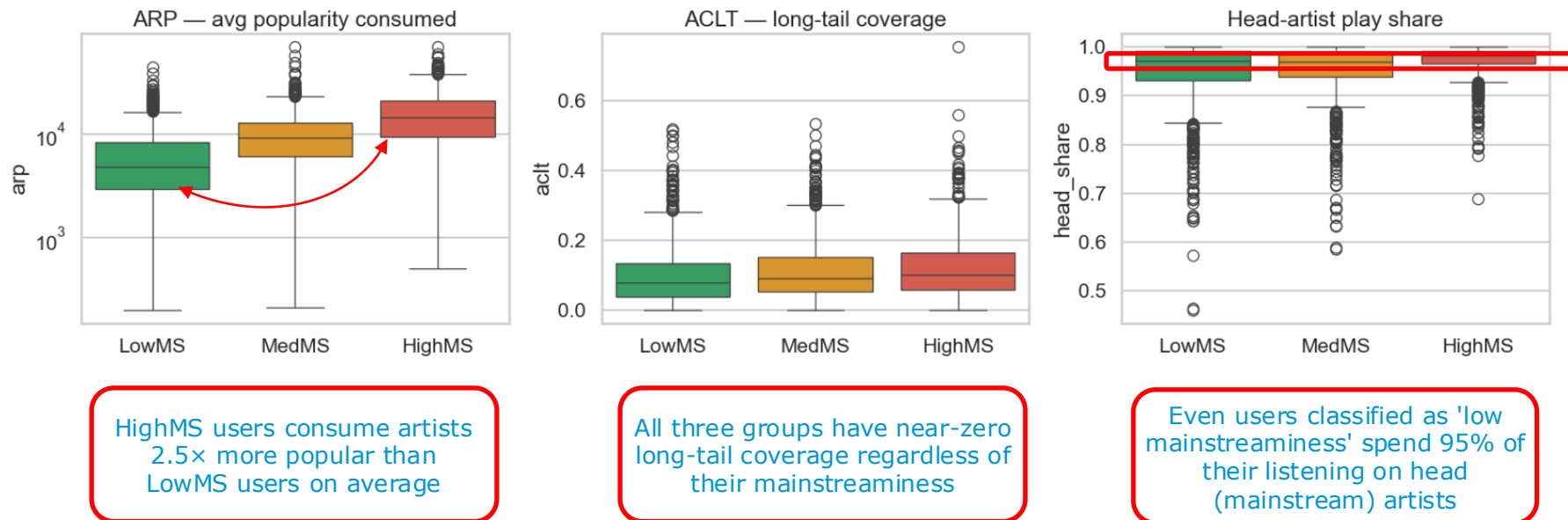


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Bias Varies Across Gender, Country, and Age

Demographic-side: popularity exposure across groups

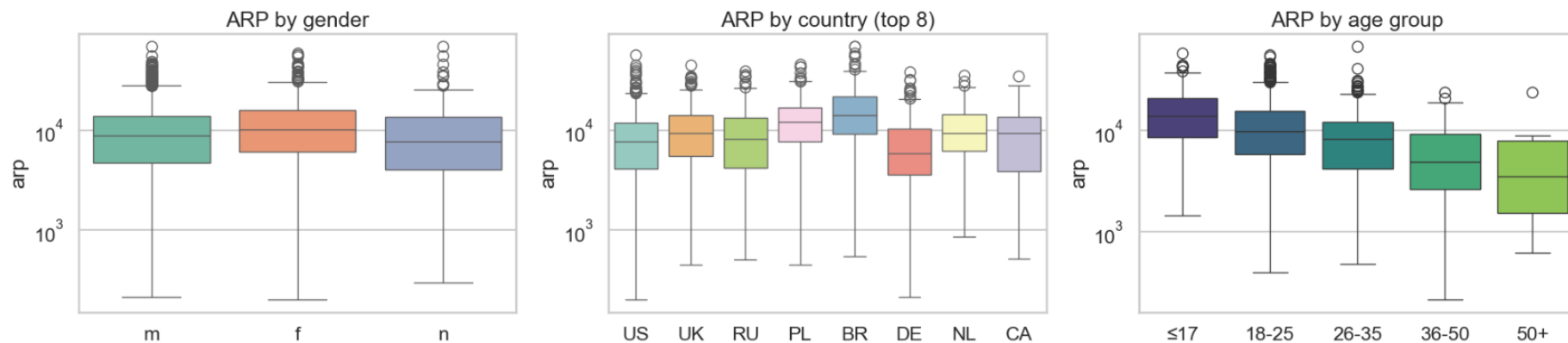


Fig 4. Measuring & Mitigating Popularity Bias in Music Recommender Systems . Section8: Demographic-side bias

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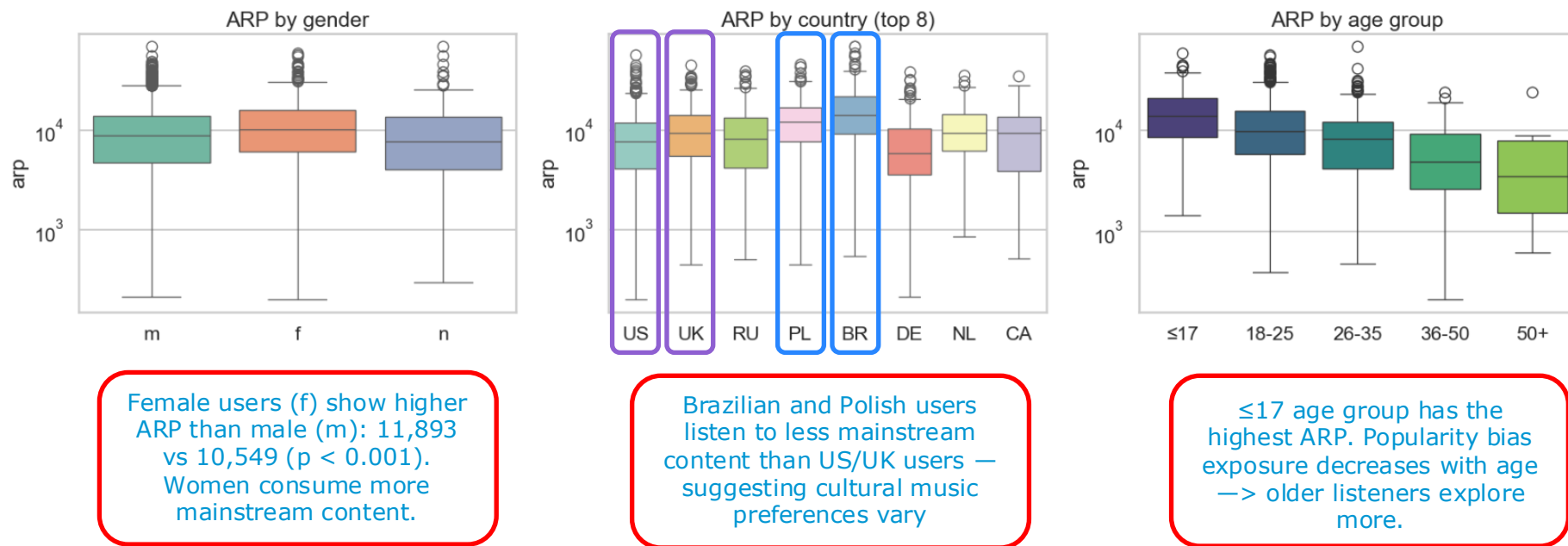


Fig 4. Measuring & Mitigating Popularity Bias in Music Recommender Systems . Section8: Demographic-side bias

Cross-dataset Validation: XITE Reproduces Bias

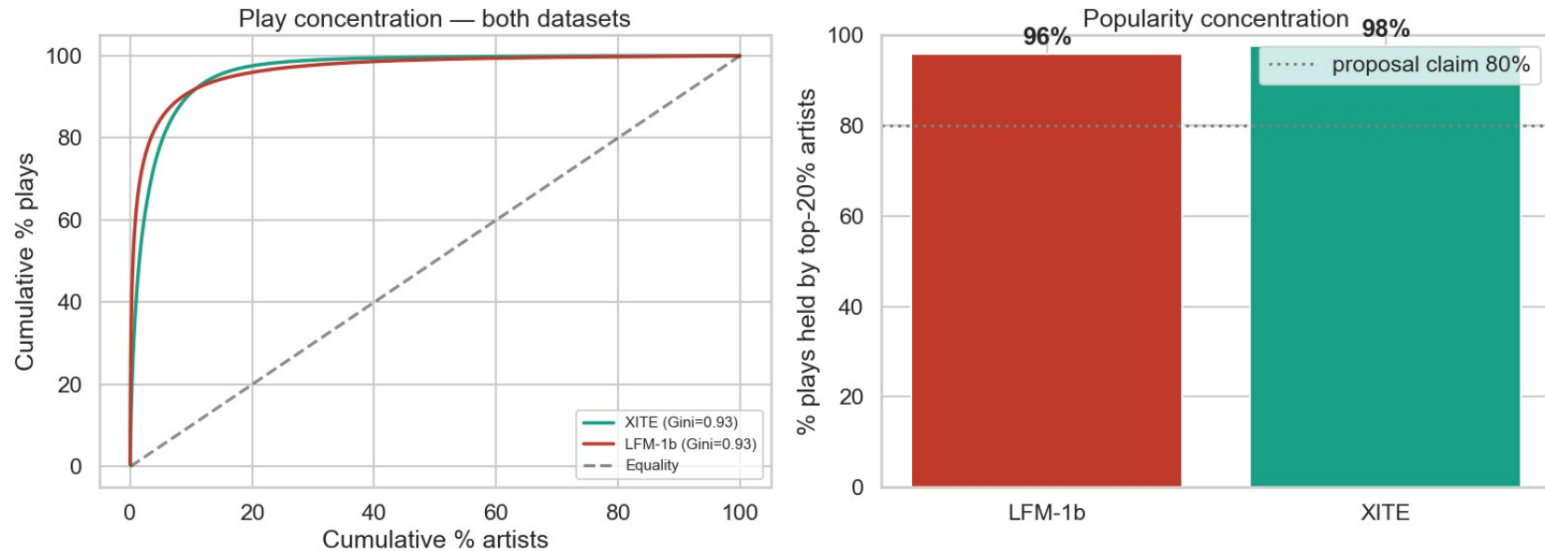
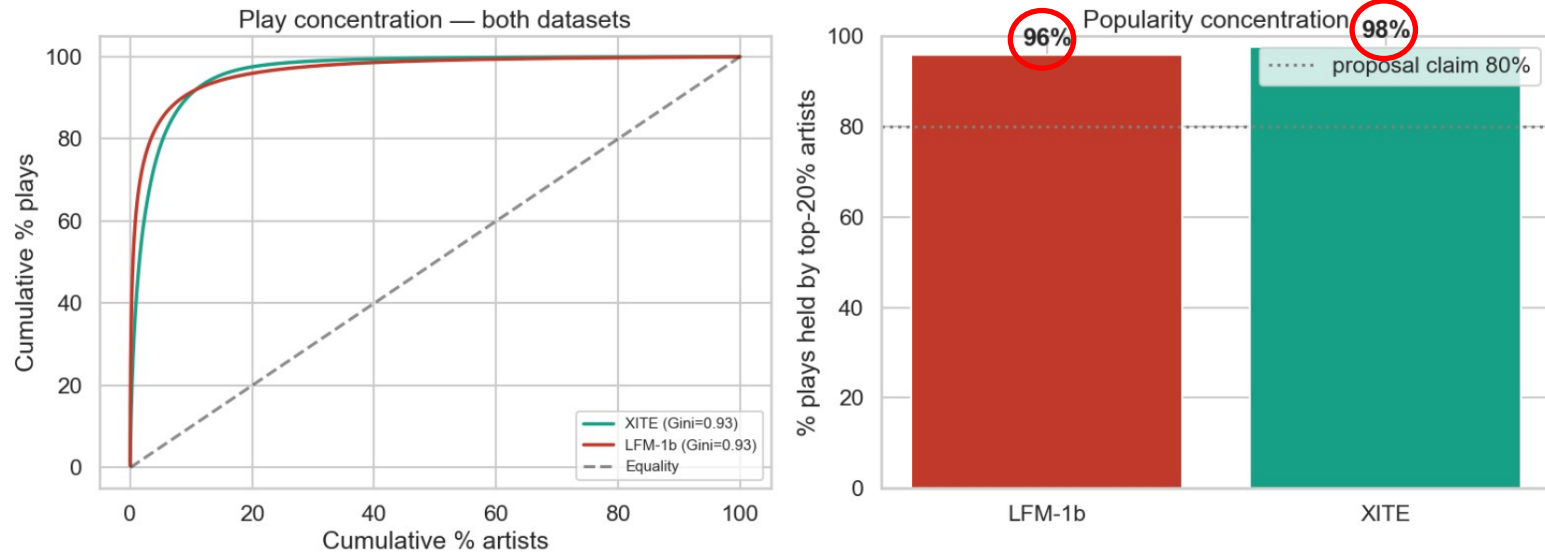


Fig 5. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section §11: Cross-dataset comparison: LFM-1b vs XITE

Cross-dataset Validation: XITE Reproduces Bias



The same structural inequality appears in scrobbling-era (Last.fm) and streaming-era (XITE) data. This is not dataset noise.

Fig 5. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section §11: Cross-dataset comparison: LFM-1b vs XITE

Top 4 Genres Capture 64% of Plays

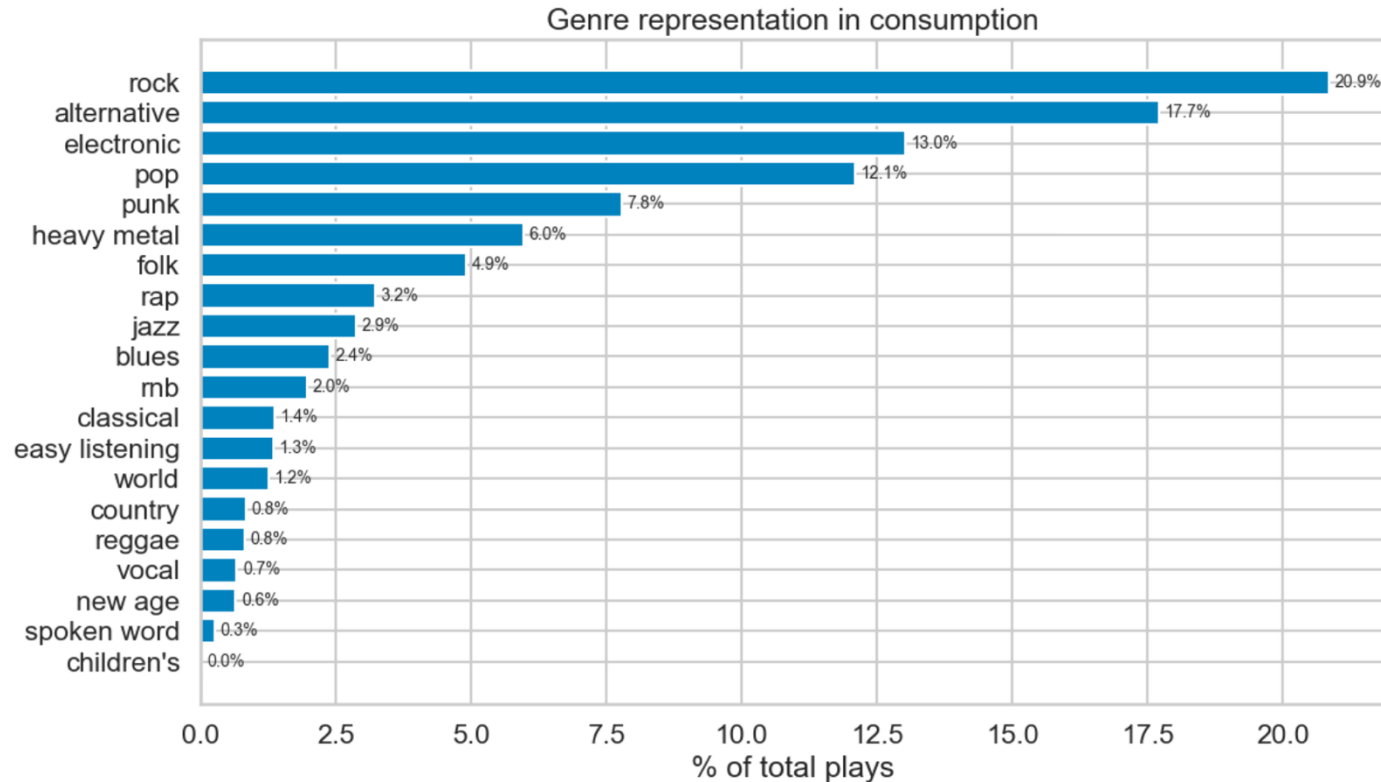


Fig 6. Measuring & Mitigating Popularity Bias in Music Recommender Systems . Section 7: Genre representation in consumption

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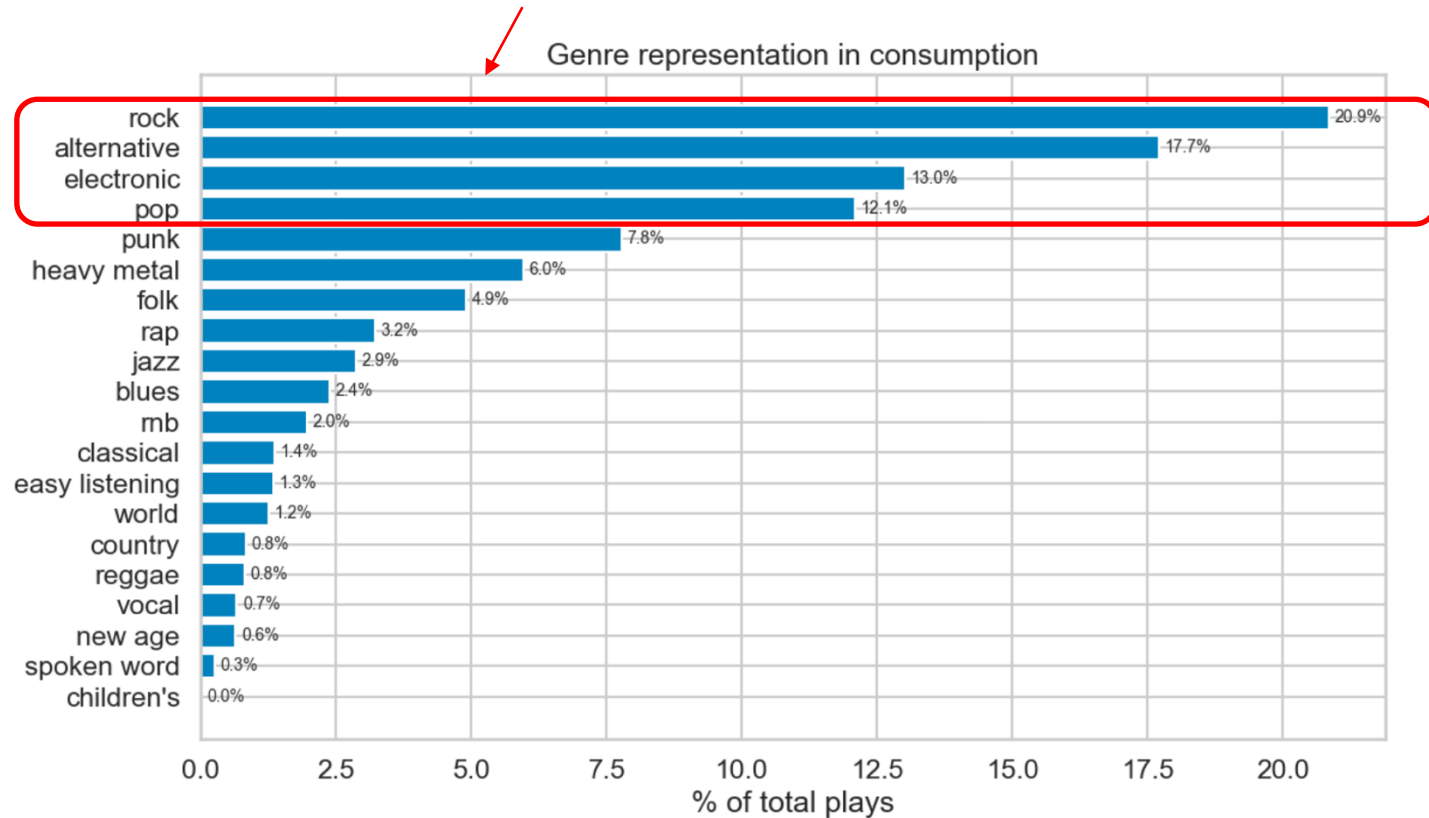


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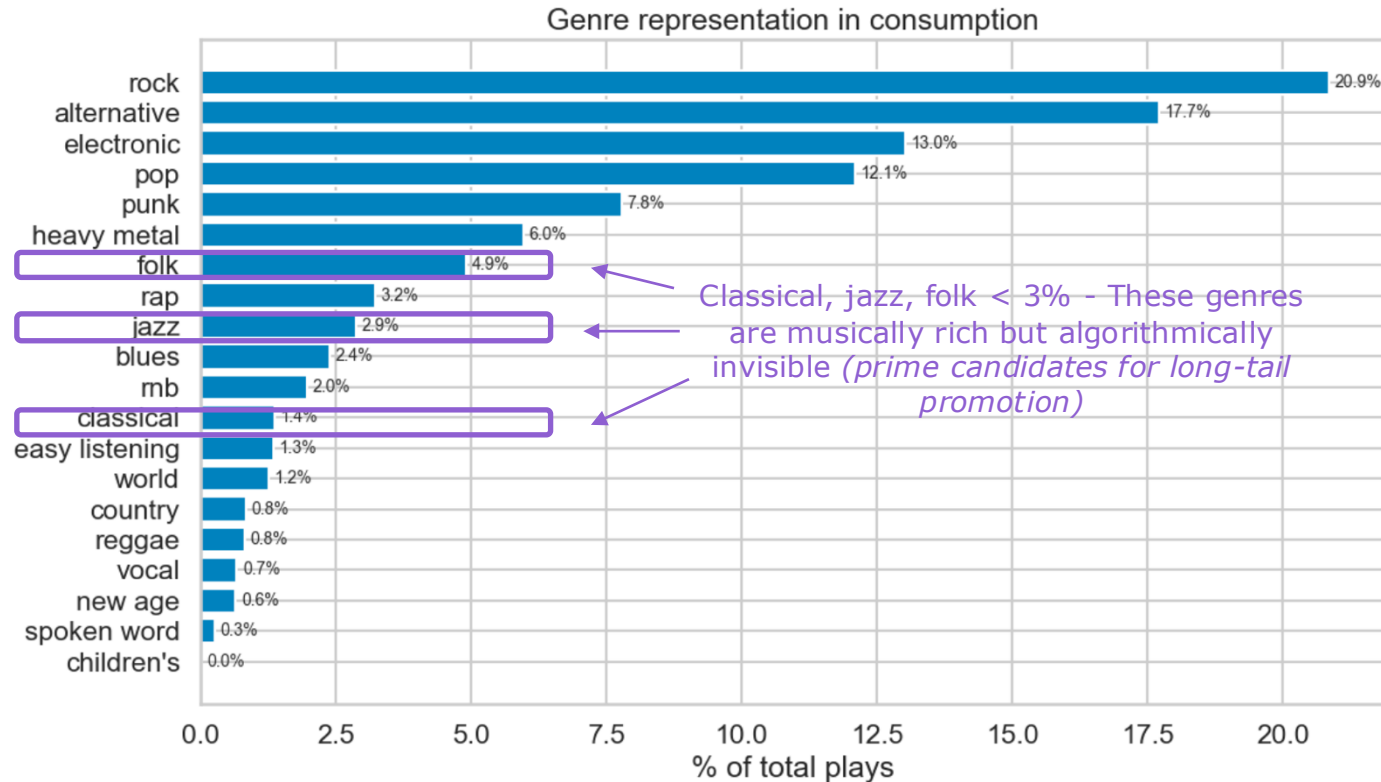


Fig 6. Measuring & Mitigating Popularity Bias in Music Recommender Systems . Section 7: Genre representation in consumption

RESULTS · RQ1 — GENRE-SIDE BIAS

All Three Models Amplify Bias Above the Raw-data Baseline

RQ1: How baseline recommenders distribute popularity

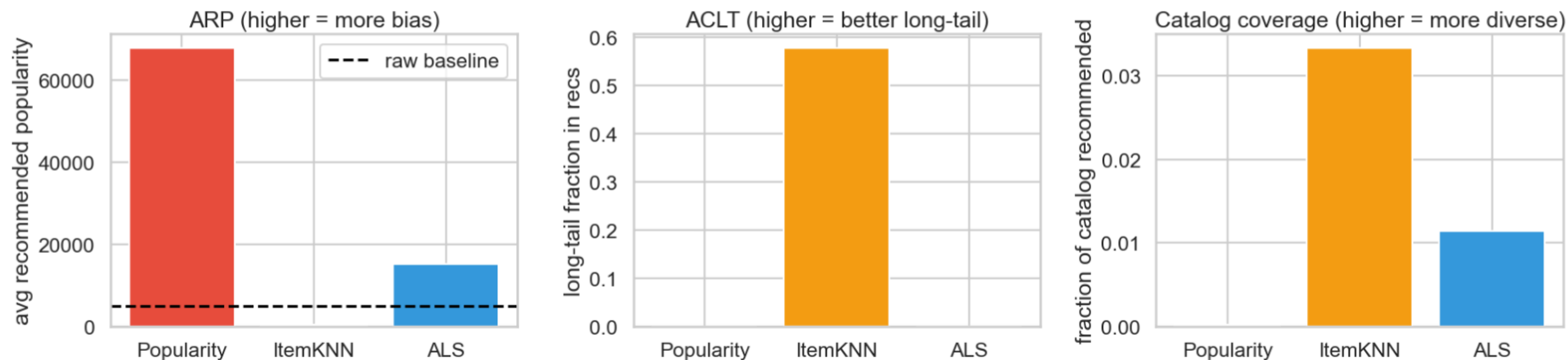
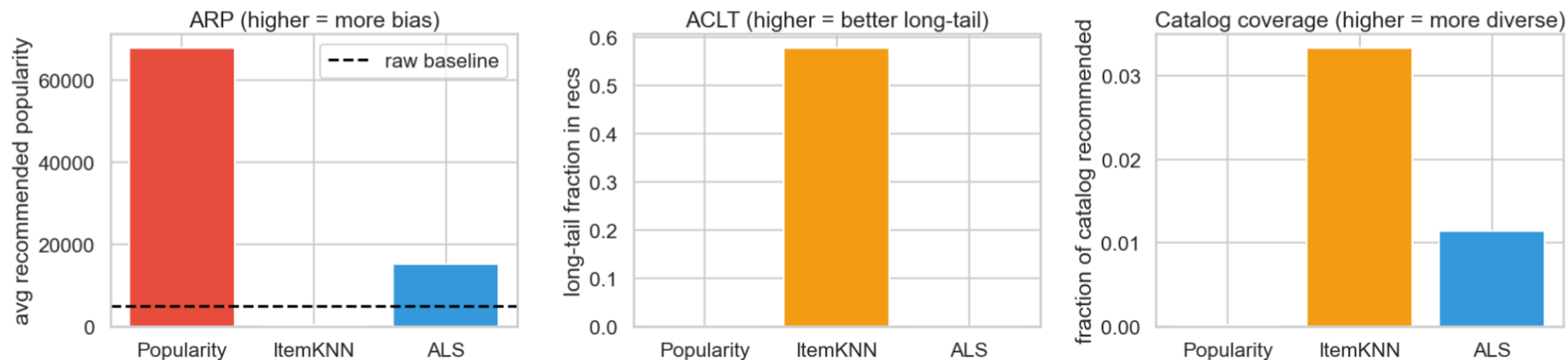


Fig 7. Measuring & Mitigating Popularity Bias in Music Recommender Systems · Section 3: Bias on Recommendations

RESULTS · RQ1 — GENRE-SIDE BIAS

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RQ1: How baseline recommenders distribute popularity



The Algorithm Makes Bias 13× Worse

PopularityRec pushes ARP to 67,739 - thirteen times above what users actually listened to.

ALS score = 0.000.

Every single recommendation goes to a mainstream artist.

96.67% of Artists Are Invisible to Every Recommender

PopularityRec recommends roughly 70 artists out of 352,805

Fig 7. Measuring & Mitigating Popularity Bias in Music Recommender Systems · Section 3: Bias on Recommendations

Re-Ranking Strategies Reduce Bias

```

week4_mitigation.py — xQuAD re-ranking
def xquad_rerank(cand_ids, cand_scores, lam, n=TOPN):
    """Greedy xQuAD: balance relevance + long-tail bonus"""
    s = np.array(cand_scores, float)
    if s.max() > s.min():
        s = (s - s.min()) / (s.max() - s.min()) # normalize
    selected, remaining = [], list(range(len(cand_ids)))
    while remaining and len(selected) < n:
        best, best_val = None, -1e9
        for j in remaining:
            tail_bonus = 1.0 if cand_ids[j] in train_tail else 0.0
            val = (1-lam)*s[j] + lam*tail_bonus # ← key formula
            if val > best_val: best_val, best = val, j
        selected.append(best); remaining.remove(best)
    return [cand_ids[j] for j in selected]

```

▲ `xquad_rerank()` — user-side re-ranking (week4_mitigation.py)

```

week4_mitigation.py — lambda sweep
lambdas = [0.0, 0.2, 0.4, 0.45, 0.5, 0.55, 0.6, 0.8]
mitigation_results = []
for lam in lambdas:
    recs_x = {ui: xquad_rerank(cand_pool[ui][0], cand_pool[ui][1], lam)
              for ui in eval_users}
    m = rec_metrics(recs_x)
    m.update({"method": "xQuAD", "param": lam,
             "NDCG@10": ndcg_at_k(recs_x, test_truth)})
    mitigation_results.append(m)
for mt in [0,2,4,6]: # min_tail sweep
    recs_f = {ui: fair_rerank(cand_pool[ui][0], cand_pool[ui][1], mt)
              for ui in eval_users}
    m = rec_metrics(recs_f)
    m.update({"method": "FairRerank", "param": mt, ...})
    mitigation_results.append(m)

```

xQuAD

User-side re-ranking

Greedy re-ranking: $val = (1-\lambda) \cdot relevance + \lambda \cdot tail_bonus$
 At $\lambda=0.45$: ACLT 0 $\rightarrow$ 0.32, NDCG 0.238 $\rightarrow$ 0.190

FairRerank

Provider-side

Guarantee min. long-tail artists per Top-N list.
 At $min_tail=6$: ACLT=0.60, Coverage=3.64%

Method	λ	NDCG	ACLT	Cov.
ALS (base)	—	0.238	0.000	1.1%
xQuAD	0.45	0.190	0.323	2.9%
FairRR	min=4	0.184	0.400	3.1%
FairRR	min=6	0.148	0.600	3.6%

Pareto Trade-off

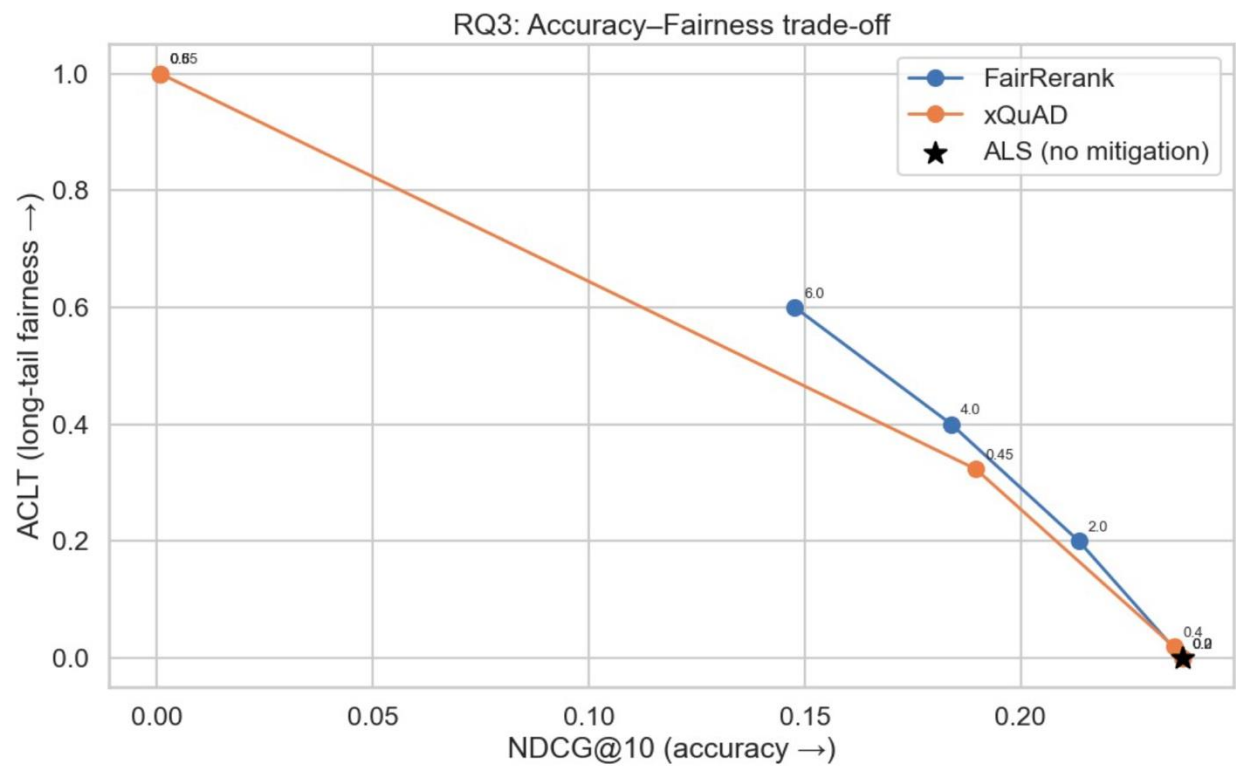


Fig 8. Measuring & Mitigating Popularity Bias in Music Recommender Systems. Section5: Trade-off & Toolkit

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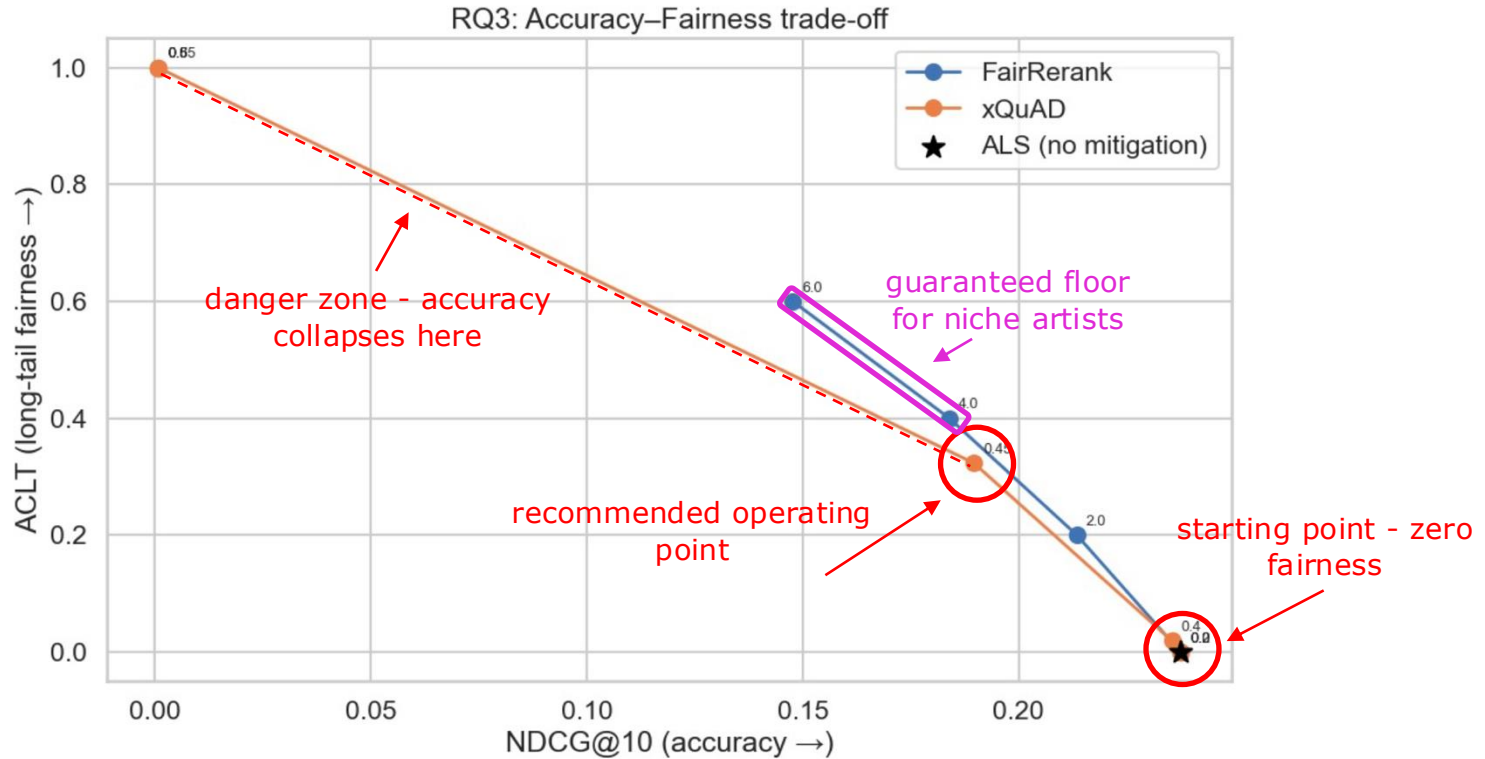


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Conclusions & Hypothesis Verdicts

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Limitations:

Data

Implicit feedback only

Play counts $\neq$ explicit ratings $\rightarrow$ a song left on repeat inflates plays, biasing all bias metrics upward.

LFM-1b is European-skewed

User base skews toward European (esp. German) listeners; findings may not generalise to other cultural contexts.

22% age data missing

Demographic age-ARP analysis is incomplete and should be treated as indicative, not conclusive.

Models & Evaluation

No hyperparameter tuning

ALS (factors=64), KNN (K=50) and λ grids were not cross-validated; optimal settings could shift the Pareto frontier.

Adversarial training not implemented

Only post-processing (xQuAD, FairRerank) done; in-processing debiasing remains future work.

Cold-start excluded

All users have training history; cold-start users (*who most need discovery*) are out of scope.

Second Dataset (XITE)

Sessions $\neq$ stable users

XITE IDs are anonymous short sessions; pseudo-user bias metrics are not directly comparable to LFM-1b.

No genre $\times$ recommender analysis

Genre bias measured in raw consumption only; how models amplify genre skew in recommendations is unquantified.